

LocalAI TV — Admin Dashboard & Database System

Final Specification Document · v1.1

Status:  Final spec after consolidating feedback from 3 senior engineers (Abishek, Sameer, Gnana) **Last updated:** 14 May 2026 **Owner:** Koneti Mohan Reddy (localaitv@gmail.com) **Replaces:** v1.0 (24 May 2026) **Document purpose:** Final shareable spec — locked and ready for build approval.

Version History

Version	Date	Changes
v1.0	24 May 2026	Initial plan, 24 sections
v1.1	14 May 2026	Consolidated 3 engineering reviews. Added production reliability, observability, security hardening, integration contract, retention policy, status enum updates.

Key changes from v1.0 → v1.1

- Queue infrastructure** added (pg_boss in Postgres) — addresses Abishek's #1, #2, #8
- Observability stack** added (Sentry, PostHog, Better Stack, OpenTelemetry) — Abishek's #9
- Security hardening** explicit (Cloudflare Turnstile, WAF, rate limits, account lockout) — Abishek's #10
- Media validation** layer (MIME, virus, NSFW, hash dedup) — Abishek's #6
- Idempotency** built into webhook contract — Sameer/Gnana
- Integration contract** locked (full Pydantic schema, HMAC, retry, cancel-job) — Sameer/Gnana
- Bulletin output structure** per Gnana's spec adopted exactly
- Retention policy** per Gnana's recommendations
- Status enum** expanded to use Gnana's failure-state vocabulary
- Production readiness checklist** added (§20a)

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1. Executive Summary

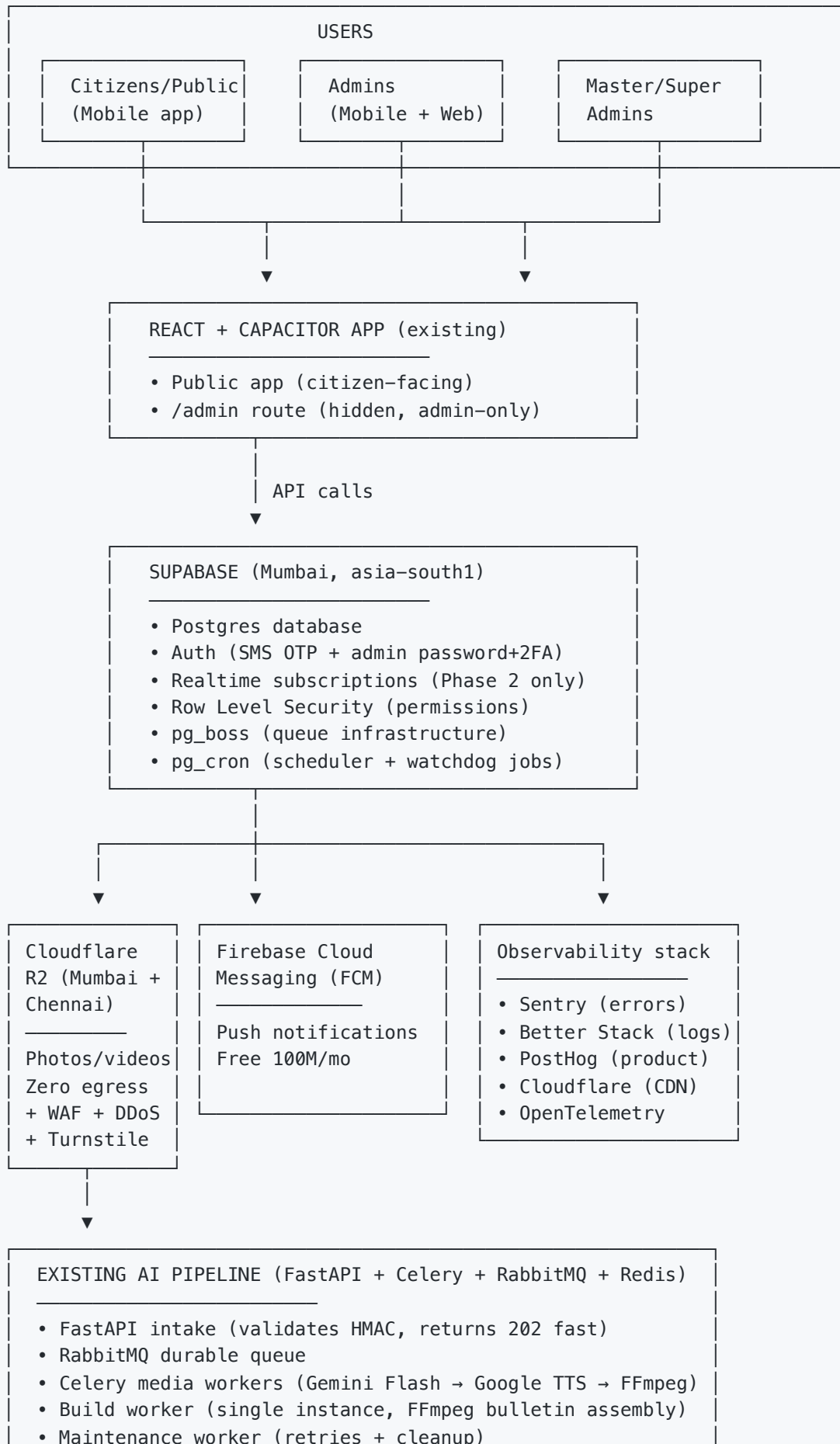
We are building an internal moderation and content management system for LocalAI TV that:

- Adds a 3-tier admin team (super → master → admin) reviewing citizen submissions before they reach the AI processing pipeline
- Integrates seamlessly with the existing **FastAPI + Celery + RabbitMQ + Redis** AI Pipeline (already production-grade per AI team's review)
- Supports state-restricted admins (Telugu admin handles AP/TG; Kannada admin handles Karnataka; etc.)
- Integrates Google NotebookLM-generated content (5 content types) as a separate auto-approved stream
- Provides time-slot-based publishing automation with notifications to uploaders
- Includes production-grade queue infrastructure, observability, and security hardening

Key non-disruption principle: The existing AI processing pipeline stays 100% unchanged. The admin layer inserts BEFORE the pipeline — only approved content reaches AI processing.

Monthly infrastructure cost (at launch): ₹400–2,500 **Team requirements beyond founder:** Existing AI team (unchanged) + Claude (Anthropic) for build **DevOps engineer needed:** No **Full-stack developer needed:** No (Claude builds everything)

2. System Architecture Overview



- Output to ai-processed/ in R2
- Callback to Admin via /api/webhook/ai-processed

3. Three-Tier Admin System

3.1 Roles

Role	Description
Super Admin	Founder-level. Full access. Only 1 person.
Master Admin	Senior moderator. Can create admins. Full state access.
Admin	Front-line moderator. Restricted to specific assigned state(s).
Citizen / Public	Unified role. No distinction between "citizen reporter" and "public".

3.2 Permissions Matrix

Capability	Super	Master	Admin
Create Master Admin	✓	✗	✗
Create Admin	✓	✓	✗
Suspend admins below tier	✓	✓ (admins only)	✗
Geographic access	All	All	Assigned states only
Upload (national/state/district/constituency)	✓	✓	✓ (within assigned states)
Bypass review (own uploads auto-approved)	✓	✓	✓
NotebookLM upload (5 sub-types)	✓	✓	✓
Approve / Modify+Approve / Reject	✓	✓	✓
Escalate to higher tier	n/a (top)	→ Super	→ Master
View all-India analytics	✓	✓	✗
Cancel approved jobs	✓	✓	✗
Audit log access	All	Team only	Own only

3.3 State Restriction (Multi-Language Support)

Admins are assigned one or more states. Queue auto-filters; RLS enforces at DB level.

Admin: Ravi Kumar

└ Phone: +91 98765 43210

└ Language access (multi-select):

- ☒ Andhra Pradesh (Telugu)
 - ☒ Telangana (Telugu)
 - ☐ Karnataka (Kannada)
 - ☐ Tamil Nadu (Tamil)
-

4. Content Workflow (End-to-End)

STEP 1 – UPLOAD

Citizen submits: 1–3 media files + headline + description + location

- Files → Cloudflare R2 (Cloudflare Turnstile blocks bots)
- ClamAV virus scan via Edge Function
- SHA-256 hash dedup
- NSFW check via Google Cloud Vision (flag for priority review)
- Metadata → Supabase (status: pending)



STEP 2 – ADMIN CLAIM (atomic lock)

Admin clicks item:

```
UPDATE content SET claimed_by=$me, claimed_at=NOW()  
WHERE id=$id AND status='pending'  
AND (claimed_by IS NULL OR claimed_at < NOW() - INTERVAL '30 min')  
RETURNING *;
```

- If 1 row returned: this admin owns it
- If 0 rows: another admin already claimed → "already claimed by X"



STEP 3 – DECISION

- ✅ APPROVE
- ✏️ MODIFY + APPROVE (original preserved read-only)
- ❌ REJECT (with reason; citizen notified)
- ESCALATE (admin → master → super)

(only if approved)



STEP 4 – WEBHOOK DELIVERY VIA QUEUE

Admin enqueues webhook job in pg_boss
Worker picks up → POST to AI Pipeline with HMAC + idempotency_key
Retry policy: 1s → 5s → 30s → 5min → 30min
Dead-letter after 5 attempts → Sentry alert + super admin notify



STEP 5 – AI PIPELINE (existing FastAPI + Celery + RabbitMQ + Redis)

FastAPI validates payload + HMAC
Returns 202 Accepted within 200ms
Enqueues Celery task with idempotency_key as task_id
Worker: Gemini Flash → Google TTS → FFmpeg
Output stored: ai-processed/<state>/<dist>/<const>/<cat>/<id>/



STEP 6 – CALLBACK TO ADMIN

```
POST /api/webhook/ai-processed
Body: { content_id, ai_output_url, thumbnail_url, status, metrics }
Admin marks content: status = 'ready_for_bulletin'
Retry 5 times if Admin unreachable; held in AI's local DLQ otherwise
```



STEP 7 – TIME-SLOT SCHEDULING

```
pg_cron + pg_try_advisory_xact_lock(slot_id)
Catch-up job for missed slots
Birthday wishes auto-scheduled for birthday date
Other content distributed across 4-5 daily slots
10 min before slot: push notification to uploader
```



STEP 8 – BROADCAST

```
Bulletin assembled (your existing 10-15min compilation)
Stored: ai-processed-bulletins/<state>/<dist>/<const>/bul_<TS>/
Live TV broadcast
Audit log records aired_at
```

5. The 8 Upload Forms — Field Audit

Note: Final field-by-field audit will be done during Phase 1 against the canonical app source (`286.App_v3.jsx`). Megan (form engineer) has independently audited the live forms and will align them per founder's decisions. The JSONB `form_data` field handles any form-specific fields without requiring schema changes.

Common rules across ALL forms:

- **Media:** Maximum 3 files (photos/videos in any combination)
 - Exception: Cars max 5, Rentals max 10 (subject to founder decision)
- **Single-language fields:** Headline / description are ONE field each (Telugu OR English; auto-detected)
- **Voice input** supported
- **Required confirmation:** "I confirm content is original" checkbox

A. 📰 News

- Media (1-3, required)
- Headline (single, required, 4-30 words)
- Description (single, required, 20-300 words)
- Location (auto-filled with constituency)

B. 🎂 Birthdays

- Recipient: name, DOB, age (auto), photos (max 3, first required)
- Up to 3 wishers: name + relation (one required), photo (optional)
- Auto-scheduled for birthday date

C. 🏠 Marriage Day (Anniversary)

- Couple name (combined field, required)
- Anniversary date
- Years auto-calculated
- Couple photos (first required)
- Up to 3 senders
- Greeting message (optional — to be ADDED to form per docs)

D. 💍 Upcoming Marriage

- Bride name, Groom name (required)
- Marriage date, Venue, Muhurtham/Reception time
- Invitation card image, Invitation video
- Family contact phone (required)

E. 📅 Events

- Event name, Event type (dropdown), Date, Time, Venue
- Description (required), Organizer name (required)
- Contact phone (required)
- Photos (max 3)

F. 🛍 Shopping

- Shop name OR Item name (one required — founder decision pending)
- Category (dropdown), Price (required), Description
- Advertisement mode (Text / Voice)
- Media (Image / Video / Audio, max 3)
- Seller phone, Location

G. 🚗 Cars / Vehicles

- Vehicle type, Brand, Model, Year
- KM driven, Fuel type (including Hybrid), Price, Owner type
- Color, Description, Seller details, Address
- Photos (max 5 — special exception)

H. 🏠 Rentals

- Property type, Configuration (1BHK/2BHK/etc), Furnished status
- Monthly rent, Security deposit (optional)

- Area, Description, Owner name + contact
- Photos (max 10 — special exception)

I. Jobs

- Job title, Company name, Job type (FT/PT/Contract/Daily wage/Internship/Fresher)
- Monthly salary (optional), Location
- Qualification (required), Skills (optional)
- Job description (required), No. of posts
- Contact phone OR email (at least one required)
- Last date to apply (optional)

Future-proofing

When adding new categories (e.g. "Who's Who"): just add to category enum + define fields. Database `form_data` JSONB handles them automatically.

6. NotebookLM Integration

5 content streams from Google NotebookLM — admin uploads, auto-approved, bypasses AI pipeline:

Source type	Description	Duration	Geographic level
Individual news	Constituency-specific items	30 sec – 1 min	Constituency
District bulletin	4-5 news compiled	4-5 min	District
State bulletin	State-wide bulletin	4-5 min	State
National bulletin	National bulletin	4-5 min	National
Debate (any level)	Discussion format	Variable	All 4 levels

Upload: Only Super/Master/Admin (admin within their states only). Auto-approved. Direct-publish lane.
`processing_mode = 'notebooklm_generated'`.

7. Database Architecture

Tech: Supabase (managed PostgreSQL) in Mumbai region.

7.1 Updated Tables

`users`

```

CREATE TABLE users (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  phone       TEXT UNIQUE NOT NULL,
  name        TEXT,
  email       TEXT,
  role        user_role NOT NULL,
  assigned_states TEXT[],
  password_hash TEXT,
  profile_photo_url TEXT,
  constituency TEXT,
  is_verified BOOLEAN DEFAULT false,
  created_by  UUID REFERENCES users(id),
  suspended   BOOLEAN DEFAULT false,
  -- Security hardening (Abishek's input)
  failed_login_count INT DEFAULT 0,
  locked_until TIMESTAMPTZ,
  last_login_at TIMESTAMPTZ,
  last_login_ip INET,
  created_at    TIMESTAMPTZ DEFAULT NOW()
);

CREATE TYPE user_role AS ENUM (
  'super_admin', 'master_admin', 'admin', 'citizen'
);

```

content (updated with v1.1 columns)

```

CREATE TABLE content (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  author_id         UUID REFERENCES users(id),
  author_role       user_role NOT NULL,
  content_source     content_source NOT NULL,
  processing_mode    processing_mode_enum NOT NULL, -- NEW from Gnana
  geography_level    geography_level NOT NULL,
  state             TEXT,
  district          TEXT,
  constituency       TEXT,
  category           content_category NOT NULL,
  target_channel     TEXT, -- NEW for TTS routing
  form_data          JSONB NOT NULL,
  media_urls         JSONB,
  media_hashes       TEXT[], -- NEW SHA-256 for dedup
  nsfw_score         FLOAT, -- NEW from Google Vision
  original_data      JSONB,
  status            content_status NOT NULL DEFAULT 'pending',
  approved_version   INT DEFAULT 1, -- NEW from Gnana
  idempotency_key    TEXT UNIQUE, -- NEW from Gnana
  claimed_by         UUID REFERENCES users(id),
  claimed_at         TIMESTAMPTZ,
  reviewed_by        UUID REFERENCES users(id),
  reviewed_at        TIMESTAMPTZ,
  rejection_reason   TEXT,
  escalated_to       UUID REFERENCES users(id),
  webhook_sent_at    TIMESTAMPTZ, -- NEW prevent double-send
  ai_job_id          TEXT, -- NEW Celery task id
  ai_callback_at     TIMESTAMPTZ, -- NEW
  ai_output_url      TEXT, -- NEW
  ai_thumbnail_url   TEXT, -- NEW
  ai_duration_seconds INT,
  ai_metrics         JSONB, -- NEW timing metrics
  created_at         TIMESTAMPTZ DEFAULT NOW(),
  updated_at         TIMESTAMPTZ DEFAULT NOW(),
  deleted_at         TIMESTAMPTZ
);

CREATE TYPE content_source AS ENUM (
  'citizen', 'admin_direct',
  'notebooklm_news', 'notebooklm_bulletin', 'notebooklm_debate'
);

CREATE TYPE processing_mode_enum AS ENUM (
  'ai_pipeline', -- needs Gemini/TTS/FFmpeg
  'raw_video', -- admin direct (skip AI)
  'notebooklm_generated' -- NotebookLM (skip AI)
);

CREATE TYPE geography_level AS ENUM ('national', 'state', 'district', 'constituency');

CREATE TYPE content_category AS ENUM (
  'news', 'birthday', 'marriage_day', 'upcoming_marriage',
  'event', 'shopping', 'car_sale', 'rental', 'job'
);

```

```
-- Updated status enum per Gnana's recommendation
CREATE TYPE content_status AS ENUM (
  'pending', 'in_review', 'rejected', 'escalated',
  'approved', 'queued', 'processing',
  'failed_retryable', 'failed_final',
  'ready_for_bulletin', 'scheduled', 'published',
  'cancelled', 'callback_lost', 'archived'
);
```

NEW

webhook_deliveries (queue + retry tracking)

```
CREATE TABLE webhook_deliveries (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  content_id        UUID REFERENCES content(id),
  idempotency_key   TEXT UNIQUE NOT NULL,
  endpoint_url      TEXT NOT NULL,
  payload           JSONB NOT NULL,
  attempt_count     INT DEFAULT 0,
  last_attempted_at TIMESTAMPTZ,
  last_response_status INT,
  last_response_body TEXT,
  delivered_at      TIMESTAMPTZ,
  dead_lettered_at  TIMESTAMPTZ,
  created_at        TIMESTAMPTZ DEFAULT NOW()
);
```

NEW

ai_callbacks (inbound from AI Pipeline)

```
CREATE TABLE ai_callbacks (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  content_id        UUID REFERENCES content(id),
  job_id            TEXT,
  idempotency_key   TEXT,
  status            TEXT,
  ai_output_url     TEXT,
  thumbnail_url     TEXT,
  duration_seconds  INT,
  error_message     TEXT,
  metrics           JSONB,
  received_at       TIMESTAMPTZ DEFAULT NOW()
);
```

content_audit_log (unchanged from v1.0)

[Same as v1.0 — see §7 of original plan]

publishing_queue (unchanged from v1.0)

[Same as v1.0]

notifications (unchanged)

[Same as v1.0]

7.2 Indexes (updated)

```
-- Fast admin queue queries (state-filtered)
CREATE INDEX idx_content_queue ON content(status, state, created_at)
  WHERE status IN ('pending', 'in_review') AND deleted_at IS NULL;

CREATE INDEX idx_content_geo ON content(state, district, constituency);

CREATE INDEX idx_approved_pool ON content(constituency, category, created_at DESC)
  WHERE status IN ('approved', 'ready_for_bulletin');

CREATE INDEX idx_reporter_stats ON content(author_id, status, created_at);

CREATE INDEX idx_publishing_pending ON publishing_queue(scheduled_for_date, scheduled_for_s
  WHERE status = 'queued';

-- For media dedup
CREATE INDEX idx_media_hashes ON content USING GIN (media_hashes);

-- For form-specific JSONB search
CREATE INDEX idx_form_data_gin ON content USING GIN (form_data);

-- For webhook delivery monitoring
CREATE INDEX idx_webhook_pending ON webhook_deliveries(content_id, attempt_count)
  WHERE delivered_at IS NULL AND dead_lettered_at IS NULL;
```

7.3 Row Level Security (unchanged from v1.0 — see §16)

8. Storage Folder Structure (updated per Gnana)

Cloudflare R2 bucket: **localaitv-storage**

```
localaitv-storage/
├── citizen-uploads/
│   ├── pending/           ← awaiting admin review
│   └── approved/         ← admin-approved, queued for AI
├── admin-direct-uploads/ ← processing_mode='raw_video'
├── notebooklm/
│   ├── individual-news/<state>/<district>/<constituency>/
│   ├── bulletins/
│   │   ├── district/<state>/<district>/
│   │   ├── state/<state>/
│   │   └── national/
│   └── debates/<level>/<state>/<district>/<constituency>/
├── ai-processed/          ← individual AI-processed items
│   ├── <state>/<district>/<constituency>/<category>/<content_id>/
│   │   ├── final.mp4
│   │   ├── headline.mp3
│   │   ├── thumb.jpg
│   │   └── meta.json
├── ai-processed-bulletins/ ← per Gnana's exact spec
│   ├── <state>/<district>/<constituency>/bul_<YYYYMMDD_HHMMSS>/
│   │   ├── final_bulletin.mp4
│   │   ├── bulletin_manifest.json
│   │   ├── thumbnail.jpg
│   │   ├── segments/
│   │   │   ├── item_001.mp4
│   │   │   └── item_NNN.mp4
│   └── metadata.json
```

R2 Lifecycle Rules (per Gnana's retention policy)

```
rules:
- prefix: ai-processed-bulletins/*/segments/
  expire_days: 1          # 24 hours
- prefix: ai-processed-bulletins/*/final_bulletin.mp4
  expire_days: 2          # 48 hours
- prefix: ai-processed-bulletins/*/thumbnail.jpg
  expire_days: 2          # 48 hours
- prefix: ai-processed-bulletins/*/bulletin_manifest.json
  expire_days: 30         # audit
- prefix: ai-processed-bulletins/*/metadata.json
  expire_days: 30         # audit
- prefix: temp/*
  expire_days: 1          # 24 hours
```

9. Hidden Admin Access

- **Web:** `https://admin.localaitv.com` (subdomain, not linked publicly)
 - **Mobile:** Hidden 5-tap on Profile reveals "Staff Sign In" OR memorize URL
 - **Login:** Phone/email + password + SMS OTP 2FA via 2Factor.in
 - **Session:** 24h expiry
 - **Failed logins:** Account locked after 5 failures, IP-blocked after 20/hour
-

10. API Endpoints (updated with v1.1 additions)

Auth (admin)

- `POST /api/admin/login` — phone+password → OTP challenge
- `POST /api/admin/verify-otp` — confirm → session
- `POST /api/admin/logout`
- `GET /api/admin/me`
- `POST /api/admin/change-password`

Content queue

- `GET /api/admin/queue?state=&category=&geo_level=`
- `GET /api/admin/content/:id`
- `POST /api/admin/claim/:id`
- `POST /api/admin/release/:id`
- `POST /api/admin/approve/:id`
- `POST /api/admin/modify-approve/:id`
- `POST /api/admin/reject/:id`
- `POST /api/admin/escalate/:id`

Admin uploads

- `POST /api/admin/upload` (auto-approved, AI pipeline)
- `POST /api/admin/upload-notebooklm` (auto-approved, direct publish)

Citizen

- `POST /api/citizen/upload`
- `GET /api/citizen/my-uploads`
- `GET /api/citizen/notifications`

User management

- `GET /api/admin/users?role=`
- `POST /api/admin/users`
- `PATCH /api/admin/users/:id`

- `DELETE /api/admin/users/:id`

Analytics

- `GET /api/admin/analytics/dashboard?from=&to=&state=&category=`
- `GET /api/admin/analytics/breakdown?dimension=`
- `GET /api/admin/analytics/reporter/:id`
- `GET /api/admin/analytics/items?...`

Publishing

- `GET /api/admin/publishing/queue`
- `POST /api/admin/publishing/schedule`
- `POST /api/admin/publishing/cancel/:id`



Webhook + Cancel job

- `POST /webhook/content-approved` (Admin → AI Pipeline)
- `POST /api/webhook/ai-processed` (AI Pipeline → Admin)
- `DELETE /webhook/cancel-job/:job_id` (Admin → AI Pipeline) 
- `GET /api/admin/job-status/:job_id` (Admin polls AI) 

Audit

- `GET /api/admin/audit?content_id=`
- `GET /api/admin/audit/recent`

11. Admin Dashboard UI

(Unchanged from v1.0 — see §11 of original plan)

12. Analytics Dashboard with Drill-Down

(Unchanged from v1.0 — see §12)

13. Channel Publishing Automation & Time Slots

Daily slots (configurable):

08:50 – 09:00	Morning
11:50 – 12:00	Lunch
16:50 – 17:00	Evening
20:50 – 21:00	Prime time
22:50 – 23:00	Late night

Scheduling rules:

- **Birthday wishes:** Auto-scheduled to closest slot to 9 AM on EXACT birthday date
- **Marriage anniversaries:** Same
- **Upcoming marriage:** Scheduled 7d, 3d, and on-day
- **Events:** Scheduled 2d, 1d, and on-day
- **NotebookLM:** Routed to appropriate-level slot
- **General news:** Round-robin distribution

Slot held vs released (per Gnana):

- Slot HELD if AI processing/callback fails
- Show "Standby filler" instead of empty broadcast
- Super admin can force-release

Pre-broadcast notification:

10 min before each slot → push to all uploaders whose content airs.

14. Push Notification Templates

(Unchanged — see §14 of original plan)

15. AI Pipeline Webhook Integration (UPDATED v1.1)

15.1 Architecture (per Sameer + Gnana's contributions)



15.2 Webhook Payload Schema (Pydantic)

```
class MediaItem(BaseModel):
    type: Literal["photo", "video"]
    url: str # 1-hour signed URL
    size_bytes: int
    duration_seconds: Optional[int] = None
    thumb_url: Optional[str] = None
    sha256: str

class ApprovedContentWebhook(BaseModel):
    content_id: str # UUID v4
    approved_version: int = 1
    idempotency_key: str # = content_id + "_v" + approved_version
    job_id: Optional[str] = None

    geography_level: Literal["national", "state", "district", "constituency"]
    state: Optional[str] = None
    district: Optional[str] = None
    constituency: Optional[str] = None

    category: Literal["news", "birthday", "marriage_day", "upcoming_marriage",
                      "event", "shopping", "car_sale", "rental", "job"]
    headline: str = Field(min_length=4, max_length=300)
    description: str = Field(min_length=20, max_length=3000)
    location: Optional[str] = None
    form_data: dict = Field(default_factory=dict)

    media_urls: list[MediaItem] = Field(min_length=1, max_length=3)

    author_id: str
    author_role: Literal["citizen", "admin", "master_admin", "super_admin"]
    approved_by: str
    approved_at: datetime

    target_channel: str # e.g. "kurnool-tv" - selects TTS_PROVIDER_<CHANNEL>
    processing_mode: Literal["ai_pipeline", "raw_video", "notebooklm_generated"]
```

15.3 Webhook Response (AI → Admin)

```
{
  "status": "queued",
  "ok": true,
  "content_id": "abc-123-def",
  "approved_version": 1,
  "job_id": "celery-task-id",
  "idempotency_status": "new", // | "already_queued" | "already_completed" | "rejected"
  "accepted_at": "2026-05-14T10:30:00Z",
  "estimated_completion_at": "2026-05-14T10:32:00Z"
}
```

15.4 Callback Format (AI → Admin)

```
POST /api/webhook/ai-processed
Authorization: Bearer <shared_callback_token>
X-LocalAITV-Idempotency-Key: <same>

{
  "content_id": "abc-123-def",
  "approved_version": 1,
  "job_id": "celery-task-id",
  "status": "ready_for_bulletin", // or failed_retryable/failed_final
  "ai_output_url": "https://r2.../ai-processed/AP/Kurnool/Kurnool-Urban/news/abc-123/final.",
  "thumbnail_url": "https://r2.../thumb.jpg",
  "duration_seconds": 47,
  "processed_at": "2026-05-14T10:35:00Z",
  "error_message": null,
  "metrics": {
    "gemini_time_ms": 1240,
    "tts_time_ms": 3820,
    "ffmpeg_time_ms": 12500,
    "total_time_ms": 17560
  }
}
```

15.5 Cancel Job Endpoint

```
DELETE /webhook/cancel-job/:job_id
```

Possible responses:

```
200 OK { "cancelled": true, "stage": "queued" | "processing" }
410 Gone { "error": "already_complete", "ai_output_url": "..." }
404 Not Found { "error": "job_not_found" }
```

15.6 HMAC Signing

```
# Sender (Admin Dashboard)
body_bytes = json.dumps(payload, sort_keys=True).encode('utf-8')
signature = hmac.new(SHARED_SECRET, body_bytes, hashlib.sha256).hexdigest()

# Receiver (AI Pipeline) – must verify using raw body bytes
raw_body = await request.body() # NOT request.json()
expected_sig = hmac.new(SHARED_SECRET, raw_body, hashlib.sha256).hexdigest()
if not hmac.compare_digest(expected_sig, request.headers['X-LocalAITV-Signature']):
    return JSONResponse({"error": "invalid_signature"}, status_code=401)
```

15.7 Retry Policy

Failure type	Action
HTTP 5xx response	Retry: 1s → 5s → 30s → 5min → 30min
Network timeout	Retry same schedule
HTTP 4xx (validation)	DO NOT retry — write to dead-letter, alert super admin
HTTP 200 OK	Mark delivered, no further attempts
HTTP 200 with <code>idempotency_status: already_queued</code>	Mark delivered (duplicate is fine)

After 5 attempts → dead-letter table → Sentry alert + super admin notification.

16. Security & Row Level Security (unchanged from v1.0 — see §16)

16a. Security Hardening (Abishek's input)

Threat	Solution	Cost
Bot uploads / scraping	Cloudflare Turnstile on all forms	₹0
DDoS / Layer 7 attacks	Cloudflare WAF (auto with R2)	₹0
API rate limiting	Per-IP token bucket in Netlify Functions	₹0
OTP abuse	Per-phone (5/hr) + Per-IP (20/hr) on <code>/api/sms-otp-send</code>	₹0
Admin brute-force	Lockout after 5 failed logins; IP block after 20/hour	₹0
Session hijack	HttpOnly + Secure + SameSite cookies, 24h expiry	₹0
File upload abuse	Per-user daily quota (20 uploads/day default)	₹0
Compromised credentials	HIBP API on admin signup	₹0
XSS	React auto-escape + CSP headers	₹0
SQL injection	Supabase parameterized queries (built-in)	₹0
Anomaly detection (Phase 2)	PostHog event rules	₹0 (free tier)
Force-logout suspect admin	Super admin "kill session" action	₹0

All free. Just configuration + middleware.

17. Audit Log (unchanged — see §17 of v1.0)

17a. Queue Infrastructure

pg_boss (Postgres-based job queue)

```
-- Installed as Supabase extension
CREATE EXTENSION pgboss;
```

Job queues

Queue name	Purpose	Concurrency
<code>webhook_delivery</code>	Admin → AI Pipeline webhook calls	10 workers
<code>ai_callback_handler</code>	Processing AI callbacks	5 workers
<code>notification_delivery</code>	Push notifications via FCM	20 workers
<code>media_validation</code>	Virus scan, NSFW, hash dedup	10 workers
<code>analytics_refresh</code>	Materialized view refreshes	1 worker
<code>cleanup</code>	R2 lifecycle cleanup, deleted content	1 worker

Idempotency

Every job has an `idempotency_key`. `pg_boss` deduplicates automatically.

Retry policy

- 5 attempts with exponential backoff: 1s → 5s → 30s → 5min → 30min
- After exhaustion → moved to `dead_letter` queue → Sentry alert

Watchdog (pg_cron)

```
-- Runs every 5 minutes
SELECT cron.schedule('stuck-job-watchdog', '*/*5 * * * *', $$
  -- Re-queue items stuck in ai_processing > 30 min
  UPDATE content SET status = 'callback_lost'
  WHERE status = 'ai_processing' AND updated_at < NOW() - INTERVAL '30 minutes';

  -- Re-queue failed_retryable jobs older than 1 hour
  UPDATE webhook_deliveries SET attempt_count = 0
  WHERE last_response_status >= 500 AND last_attempted_at < NOW() - INTERVAL '1 hour';
$$);
```

At scale

When `pg_boss` insufficient (>10K jobs/min sustained):

- Migrate to **Cloudflare Queues** (\$0.40/million messages)
- Or self-hosted **Redis + BullMQ**

17b.

NEW

Observability & Monitoring

Tool	Purpose	Free tier	Cost @ scale
Sentry	Error tracking, stack traces, releases	5K errors/mo	₹0–2K
Better Stack (Logtail)	Log aggregation	1 GB/mo	₹0–3K
PostHog	Product analytics, session replay, feature flags	1M events/mo	₹0–25K
Cloudflare Analytics	CDN + R2 metrics, request analytics	Built-in	₹0
Supabase Logs	DB queries, slow query log	Built-in	₹0
Netlify Function Logs	Serverless function logs	Built-in	₹0
OpenTelemetry	Distributed tracing (Phase 3)	Self-hosted	₹0
Grafana Cloud	Custom dashboards (queue depth, p95 latency)	10K series	₹0
Flower (Celery monitoring)	AI Pipeline queue + worker visibility	Self-hosted	₹0

Critical alerts (PagerDuty / Slack)

Alert	Threshold
Webhook delivery dead-letter	>0 in last 15 min
AI callback missing	Content in <code>ai_processing</code> >30 min
Queue depth	>500 pending in any queue
Worker count	Below configured minimum
5xx response rate	>1% in last 5 min
Admin login failures (per IP)	>20/hour
Sentry error rate	>50/hour

18. Tech Stack Summary (updated)

Layer	Service	Cost @ launch	Cost @ 10K users
Database + Auth + Realtime	Supabase (Mumbai)	₹0	₹2,100/mo
Job queue	pg_boss (in Supabase)	₹0	₹0
Media storage	Cloudflare R2	₹0	₹3,000/mo
WAF + CAPTCHA	Cloudflare Turnstile + WAF	₹0	₹0
Push notifications	Firebase Cloud Messaging	₹0	₹0
SMS OTP	2Factor.in	₹400/mo	₹2,500/mo
AI script	Google Gemini Flash	₹500-2K	₹5,000/mo
AI voice (TTS)	Google TTS	₹1-3K	₹8,000/mo
NSFW detection	Google Cloud Vision	₹0 (free 1K/mo)	₹2-8K/mo
App hosting	Netlify	₹0	₹1,500/mo
Error tracking	Sentry	₹0	₹0
Log aggregation	Better Stack	₹0	₹0-3K
Product analytics	PostHog	₹0	₹0-25K
Email (optional)	Resend	₹0	₹0
Domain	localaitv.com	₹85/mo	₹85/mo
TOTAL		₹2-5K/mo	₹25-50K/mo

19. Cost Projections

Same as v1.0 — see original §19. Production hardening adds ₹0 at launch.

20. Implementation Phases

Phase 1 — Foundation

- Supabase project (Mumbai)
- Database schema (all tables + indexes + RLS)
- pg_boss queue setup
- Sentry + Better Stack integration
- Auth flow (password + SMS OTP)
- One super admin (founder)
- Cloudflare R2 + Turnstile + WAF setup

Phase 2 — Admin Dashboard Core

- Dashboard layout + nav
- Review queue with state-filtering
- Item detail / review screen
- Claim/lock mechanism (atomic SQL)
- Approve / Reject flows
- Webhook delivery via pg_boss queue
- Idempotency + dead-letter handling

Phase 3 — Admin Workflow Complete

- Modify & Approve (original retention)
- Escalation flow
- Admin direct upload
- Push notification integration (FCM)
- Notifications to citizens
- Media validation pipeline (MIME + virus + hash + NSFW)

Phase 4 — Channel Publishing + NotebookLM

- Broadcast slots config
- pg_cron + advisory locks scheduler
- Birthday auto-scheduling
- NotebookLM upload (5 sub-types)
- Time-slot notifications to uploaders
- Publishing queue UI

Phase 5 — Analytics + Team Management

- Analytics dashboard with drill-down
- Reporter profiles
- Date range filters
- Team management UI
- Audit log viewer
- Cancel-job UI for super admin

Phase 6 — Integration Testing + Onboarding

- AI Pipeline webhook end-to-end testing
- Load test (k6 or Artillery)
- Disaster recovery runbook
- Onboard first admins
- User documentation
- Production deployment

20a. Production Readiness Checklist

Before declaring v1 launched:

- ☐ Load test (1K concurrent admins, 10K citizen uploads/hour)
 - ☐ Smoke tests for every endpoint
 - ☐ Disaster recovery runbook documented
 - ☐ Rollback procedure tested
 - ☐ All Sentry alerts firing correctly to Slack
 - ☐ Cloudflare WAF rules tuned to actual traffic
 - ☐ Rate limits validated against real usage
 - ☐ R2 lifecycle rules applied
 - ☐ Backup strategy documented (Supabase auto-backup confirmed)
 - ☐ Admin onboarding doc tested with non-technical user
 - ☐ AI Pipeline webhook integration tested end-to-end with all 9 categories
 - ☐ Cancel-job flow tested at each stage
 - ☐ Idempotency tested (duplicate webhook sends)
 - ☐ Failover behavior tested (kill worker mid-job)
-

21. Team Requirements

Required:

- **Founder (Koneti Mohan Reddy)** — decisions, API keys, approval
- **AI Team (Sameer + Gnana)** — webhook adapter, ~1 week effort, existing pipeline unchanged
- **Claude (Anthropic)** — admin dashboard + database + integrations
- **Form Engineer (Megan)** — align live app forms with documented schema

NOT required:

-  DevOps engineer
-  Full-stack hire (Claude builds)
-  DBA (Supabase managed)

Optional later:

- Part-time React maintainer (5-10 hrs/week post-launch)
-

22. What's IN / OUT of v1

 **IN v1 (additions from v1.0)**

- All v1.0 features
- **Production-grade queue infrastructure** (pg_boss)

- **Observability stack** (Sentry + Better Stack + PostHog)
- **Security hardening** (Turnstile, WAF, rate limits, account lockout, file quotas)
- **Media validation** (MIME, virus, hash, NSFW)
- **Idempotency + HMAC + retry** in webhook contract
- **Bulletin output structure** per Gnana's spec
- **Retention policy** (R2 lifecycle rules)
- **Cancel-job flow** for super admin
- **Production readiness checklist**

✗ OUT of v1 (deferred)

- Public analytics / leaderboards
- AI-assisted moderation (auto-flag)
- Reporter compensation/payout
- Reporter ranking-based auto-approval (Phase 2 — data tracked now)
- Bulk approve/reject
- Comment moderation
- Multi-language admin UI chrome
- Realtime websocket queue (polling sufficient for v1)
- ClickHouse OLAP (Phase 2 when needed)

23. What I Need From You — Action Items

Same as v1.0 §23, plus:

Item	New in v1.1
A. Supabase	(existing)
B. Cloudflare R2	(existing) — now also enables Turnstile + WAF automatically
C. Firebase FCM	(existing)
D-G. Existing keys	(existing)
H. DNS for admin subdomain	(existing)
I. AI team webhook URL	(existing) — Sameer/Gnana to provide
J. Optional Resend	(existing)
 K. Sentry account	https://sentry.io — free tier, 1-min signup
 L. Better Stack (Logtail)	https://betterstack.com — free 1 GB/mo
 M. PostHog cloud account	https://posthog.com — free 1M events/mo

Additional founder time: ~10 min for the 3 new accounts. **Additional cost:** ₹0 (all free tiers cover v1 scale).

24. Approval & Sign-off

Founder approval

Item	Confirmed by founder?
Three-tier admin + state restrictions	☐
Form fields (Megan-aligned during Phase 1)	☐
Supabase + R2 + pg_boss + Sentry stack	☐
Webhook contract (HMAC + idempotency + retries)	☐
Bulletin output structure (Gnana's spec)	☐
Retention policy (48hr bulletins, 30d audit)	☐
Security hardening (Turnstile, WAF, rate limits)	☐
Push notifications (FCM, free)	☐
NotebookLM 5 sub-types	☐
Analytics dashboard with drill-down	☐
Implementation phased build	☐
₹2,000–5,500/mo infrastructure cost	☐
OUT of v1: payout, leaderboards, AI moderation, realtime	☐

AI Team approval

Item	Confirmed?
Webhook contract (FastAPI intake, 202 fast, async Celery)	☐
Idempotency = content_id + approved_version	☐
Cancel-job endpoint feasibility	☐
Callback retry policy (5 attempts)	☐
Bulletin output structure adoption	☐
R2 download adapter implementation (~1 day)	☐
Direct-publish lane (admin/notebooklm)	☐
OpenTelemetry instrumentation	☐
Fallback policies (Gemini → OpenAI, GCP TTS → Sarvam)	☐
Separate Celery queue for high-volume TTS	☐

Final approval to begin Phase 1:

Founder signature: _____ Date: _____

Contact

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Phase 1 build begins